BALKIS SELMI

10 Street Mali, Cité Saada, Oued Ellil, Manouba, Tunisie • +216 94 380 679 • Balkisa622@gmail.com

SUMMARY

Geomatics Engineering student passionate about leveraging geospatial technologies, artificial intelligence, and data science to address environmental and climate challenges. Driven by a strong commitment to sustainability, I develop innovative solutions for water resource management, disaster risk reduction, ecosystem restoration, and smart environmental monitoring. Through academic research, leadership in environmental initiatives, and interdisciplinary projects, I strive to transform technology into impactful solutions that build more resilient and sustainable communities.

PROFESSIONAL EXPERIENCE

1- Computer science and engineering

Data Science & AI Intern – STEG (Jun 2026)

- Developed a centralized monitoring system collecting operational data from more than 53 district solar PV stations.
- Integrated production, power generation, and inverter fault data into a single dashboard application.
- Developed an AI model to forecast photovoltaic energy production one year in advance.

Data Annotator – Doxaria (Sept 2025-Now)

- Labeled and categorized datasets to support AI/ML model development.
- Verified data accuracy and ensured consistency with project guidelines.
- Maintained high productivity and quality standards within short deadlines.

Intern – Geomatics & Remote Sensing , Entreprise Tunisienne d'Activités Pétrolières (ETAP), Tunisia (July 2025)

- Processed geospatial data and performed coordinate system transformations using Petrel and ArcGIS Pro.
- Contributed to documentation and analysis of mapping systems (IGN & RGS) used in Tunisia.
- Gained practical experience in applying geodesy and geomatics in the oil and gas sector.

Intern – Engineer for New Energy, APA Group, Australia (Feb 2025)

- Conducted a professional simulation within the Project Development team on renewable energies.

- Applied concepts related to solar photovoltaics and battery storage to draft a Basis of Design.
- Calculated installed capacity, load factor, and potential CO_2 emissions reduction of a solar power plant.
- Developed awareness of safety aspects and gained hands-on experience in troubleshooting and problem-solving during operations.

Intern – Python Developer, Code Alpha, India (Aug 2024 – Sept 2024)

- Developed and optimized Python scripts.
- Designed desktop applications using PyQt5 and other libraries.
- Collaborated on innovative solutions, strengthening technical skills and professional experience.

2- Civic activities and teaching:

Arabic Tutor – Preply (Mar 2025 – Present)

- Delivered personalized 1:1 Arabic (MSA and conversation) lessons tailored to students' goals and proficiency levels.
- Designed lesson plans covering speaking, grammar, vocabulary, reading, and pronunciation.
- Monitored student progress and adapted teaching strategies to improve fluency and confidence.

Social Media & Community Manager – Association Tunisienne des Éducateurs (Jun 2024 – Present)

- Created and published engaging content promoting educational initiatives and events.
- Managed the association's online presence and digital brand image.
- Coordinated events, workshops, partners, and communication campaigns to increase visibility and participation.

EDUCATION

Engineering Degree in Geomatics – Manouba School of Engineering

- Preparatory cycle: Sept 2023 – June 2025
- Engineering cycle: Aug 2025 – Present (Degree expected June 2028)

Secondary School Diploma – Pionner High School of Ariana (sept. 2019 – juin 2023)

- Distinction: Very Good – Score: 17.28/20

PROJECTS

- Research: “A new framework to transform education in Tunisia: SDGs as a lever for progress” – ENIM (12 Dec 2024).
- Developed and implemented a coordinate system conversion in Petrel (Carthage RGS → Carthage IGN) (1–31 July 2025).
- HydroTwin – Intelligent water management solution (25 July 2025 – Present).
- Drought Monitoring Dashboard: Interactive GIS dashboard using satellite imagery and Python to monitor drought conditions.
- HydroTwin Mobile App: Digital twin dashboard integrating GIS and AI to predict flooding and water risk zones.

LANGUAGES & SKILLS

- **Languages:** Arabic, English, French, Spanish

- **Programming & Data Science:**
  MATLAB basics, C++, Data Cleaning & Preprocessing(PostgreSQL Python(NumPy, Pandas, Matplotlib), SQL), Feature Extraction, Model Training & Evaluation, Web Development (Html, CSS, Js/ React/ Django/ PHP), Mobile Development(Androis Studio)
- **Image Processing & Analysis:**
  Supervised and Unsupervised Classification, Image Segmentation, Object Detection, Edge Detection, Histogram Equalization, Morphological Operations, Filtering & Enhancement
- **Remote Sensing & GIS:**
  QGIS, ArcGIS, ENVI, Georeferencing, Raster & Vector Analysis, Spatial Modeling, Environmental Change Detection, Geostatistics, Google Earth Engine
- **AI & Machine Learning:**
  Model training (classification & clustering), Neural Network fundamentals, Data Labeling for AI, AI-based Remote Sensing Applications
- **Other Tools & Platforms:**
  Excel, Jupyter Notebook, Google Colab, GitHub
- **Soft Skills:**
  Analytical thinking, Attention to detail, Communication, Collaboration, Problem-solving, Adaptability

ACHIEVEMENTS & AWARDS

- Top 10 National Winner – Represented Tunisia at the international finals in Dubai (2016).
- National Second Prize – Filmmaking Production, 'Creatives in the Quarantine' competition.
- Presented the HydroTwin project at the International Space Star Conference and received certification for the scientific poster presentation.
- Selected among the Top 15 national research ideas in BrainX; completed the BrainX Bootcamp organized by Open Startup and received certification.
- Selected by JCI TBS Tunisia for the TOYP 2026 (Ten Outstanding Young Persons) program and advanced to the Top 30 national finalists for the African stage.